Midterm Exam II

Due: 3 pm April 27, 2022

Name________________________________

Student ID__________________________________

VERSION A

Instructions:

- NO interactions and communications with anyone in any form—**academic integrity and honesty** is important
- Answer all questions using a blue/black pen on A4 papers
- Upload your answers and the first page of the exam to WM5 by **3:20 pm** (20 points off if uploading after 3:20 pm, and any submission after 3:30 pm will not be accepted)
- Read and sign the statement below

By signing this, I declare that the submitted exam was produced independently by me, without help from others.

Signature ____

Microeconomics **Midterm 2**, Spring 2022

Name and Student ID:

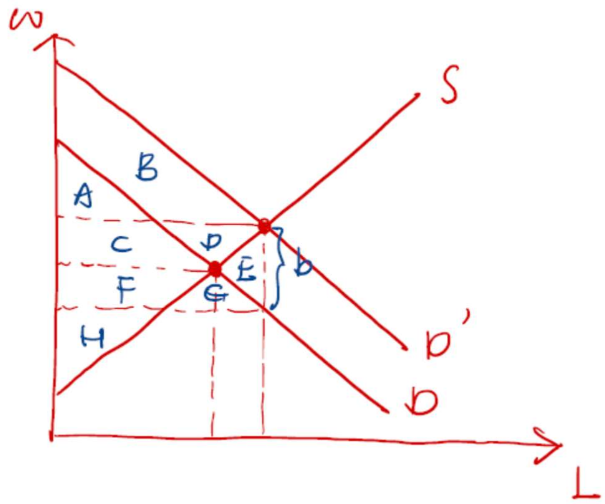
1. [20 points] (Minimum Wage and Wage Subsidy) Discuss the welfare effects of a minimum wage and a wage subsidy in a perfect competitive labor market.

    a. [10 points] The government imposes a minimum wage above the equilibrium wage. Use a graph to show how the minimum wage affects the worker surplus, the firm surplus, and the social welfare.

    b. [10 points] Another way to increase the wage workers receive is to provide a specific subsidy of *b* to firms. Use a graph to discuss the effect of the subsidy on the worker surplus, the firm surplus and the social welfare.

a. See lecture notes.

b.



	No Sub.	Sub.	Δ
CS	$A + C$	$A + C + F + G$	$F + G$
PS	$F + H$	$C + D + F + H$	$C + D$
Gov't Exp	0	$-(C + D + E + F + G)$	$-(C + D + E + F + G)$
W			$-E$

2. [10 points] (Monopsony) **True or False**: If there is monopsony in the labor market, a minimum wage law can lead to increased employment and welfare.

   See lecture notes.

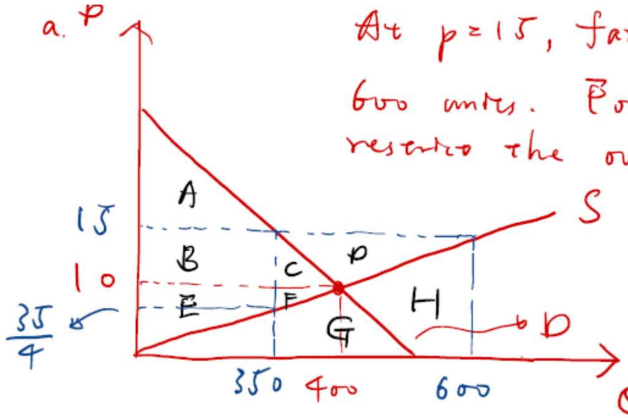
3. [20 points] (Price Support) The market demand for sorghum is given by $Q_d = 500 - 10P_d$, while the market supply curve is given by $Q_s = 40P_s$. The government would like to increase the income of farmers and is considering two alternative government interventions: an acreage limitation program and a government purchase program.

   a. [8 points] The government's goal is to increase the price of sorghum to \$15 per unit. This is the support price. How much would the government need to pay farmers in order for them to voluntarily restrict their output of sorghum to the level demanded at \$15 per unit?

   b. [8 points] As an alternative way to support a price of \$15, suppose the government purchases the difference between the quantity demanded at a price of \$15 and the quantity supplied. How much does the government spend on this price support program?

   c. [4 points] Compare the deadweight loss resulting from the acreage limitation program and the government purchase program.

a.



At p=15, farmers want to produce 600 units. For them to voluntarily restrict the output to 350, the gov't has to compensate them C+D+F, the PS farmers give up.

$C + D + F = (600 - 350) \cdot (15 - \frac{35}{4}) \cdot \frac{1}{2} = 781.25$

$\Delta CS: -(B + C)$

$\Delta PS: B + C + D \quad \Rightarrow D.W. = C + F = \frac{1}{2}(15 - \frac{35}{4}) \cdot 50$

$\Delta Exp: C + D + F \qquad\qquad = 156.25$

b. $\Delta Exp = C + D + F + G + H = 15 \times 250 = 3750$

$\Delta CS: -(B + C)$

$\Delta PS: B + C + D \quad \Rightarrow D.W. = C + F + G + H$

$= 3750 - D$

$= 3750 - \frac{1}{2} \times 5 \times 250 = 3125$

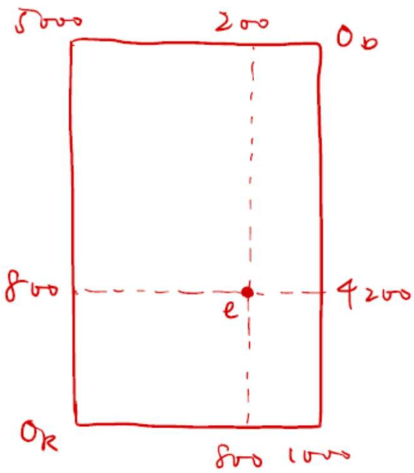
4. [20 points] (General Equilibrium) Two consumers, Ron and David, together own 1,000 baseball cards and 5,000 Pokémon cards. Let x_R denote the quantity of baseball cards owned by Ron and y_R denote the quantity of Pokémon cards owned by Ron. Similarly, let x_D denote the quantity of baseball cards owned by David and y_D denote the quantity of Pokémon cards owned by David. Suppose, further, that for Ron, $MRS_R=y_R/x_R$, while for David, $MRS_D=y_D/2x_D$. Finally, suppose $x_R = 800$, $y_R = 800$, $x_D = 200$, and $y_D = 4{,}200$.

a. [10 points] Draw an Edgeworth box that shows the set of feasible allocations and identify the endowment point in this simple economy.

b. [5 points] Show that the current allocation of cards is not economically efficient.

c. [5 points] Identify a trade of cards between David and Ron that makes both better off. (Note: There are many possible answers to this problem.)

a.



b. At the current allocation,

$$MRS^R = \frac{800}{800} = 1 \ \& \ MRS^D = \frac{4200}{2\times 200} = 10.5$$

c. Ron can give David 10 baseball cards in exchange of 20 pokémon cards. Both of them will be better off because

$$MRS^R_e < 2 < MRS^D_e$$

5. [20 points] (General Equilibrium) Answer the following two questions:

a. Sarah and Andrew are two traders in a pure exchange economic with two goods, Bikes (B) and Computers (C). Sarah's preferences are described by the Cobb-Douglas Utility function:

$$U_S = B_S^{1/3} C_S^{2/3}$$

Andrew's preferences are given by:

$$U_A = B_A^{1/2} \frac{1}{2} C_A^{1/2}$$

Assume the price of Bikes is 1 and the price of computers is p. The initial endowments are $B_A^0 = 10$, $B_S^0 = 20$, $C_A^0 = 20$ and $C_S^0 = 10$. Solve for the competitive equilibrium prices (relative prices) and quantities.

Determine the demand equations using the incomes and preferences. Sarah's income is:

$$\mathbf{20 + 10p = Y_S}$$

Her demand curves are:

$$\mathbf{B_S = Y_S/3 = (20 + 10p)/3}$$
$$\mathbf{C_S = 2Y_S/3p = (40 + 20p)/3p}$$

Andrew's income is:

$$\mathbf{10 + 20p = Y_A}$$

His demand curves are:

$$\mathbf{B_A = Y_A/2 = 5 + 10p}$$
$$\mathbf{C_A = Y_A/2p = (5 + 10p)/p}$$

Setting supply ($B_A + B_S = 30$) equal to demand:

$$\mathbf{30 = (20 + 10p)/3 + 5 + 10p}$$

Solving:

$p = 55/40$ (Equating the supply and demand for computers yields the same result.)

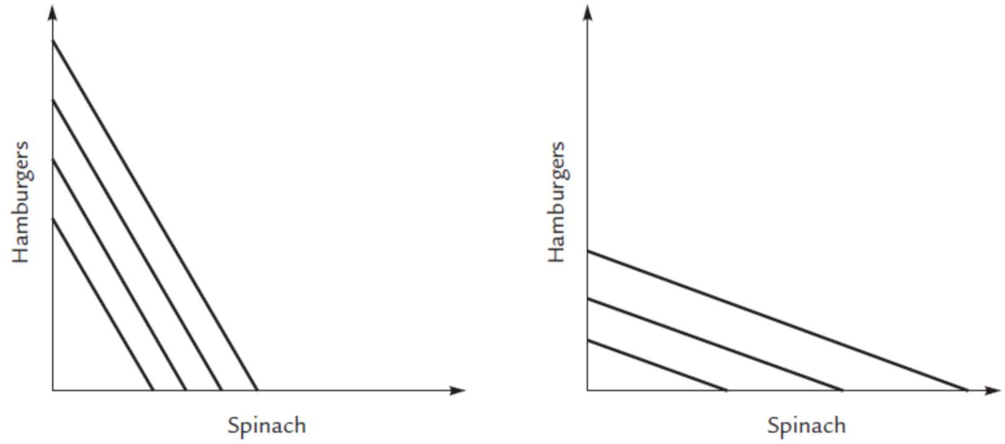
Substitute p into the demand curves:

$B_A = 5 + 10 \cdot \frac{55}{40} = 18.75 \Rightarrow B_S = 11.25$

$C_A = 18.75 / \frac{55}{40} \cong 13.64 \Rightarrow C_S \cong 16.36$

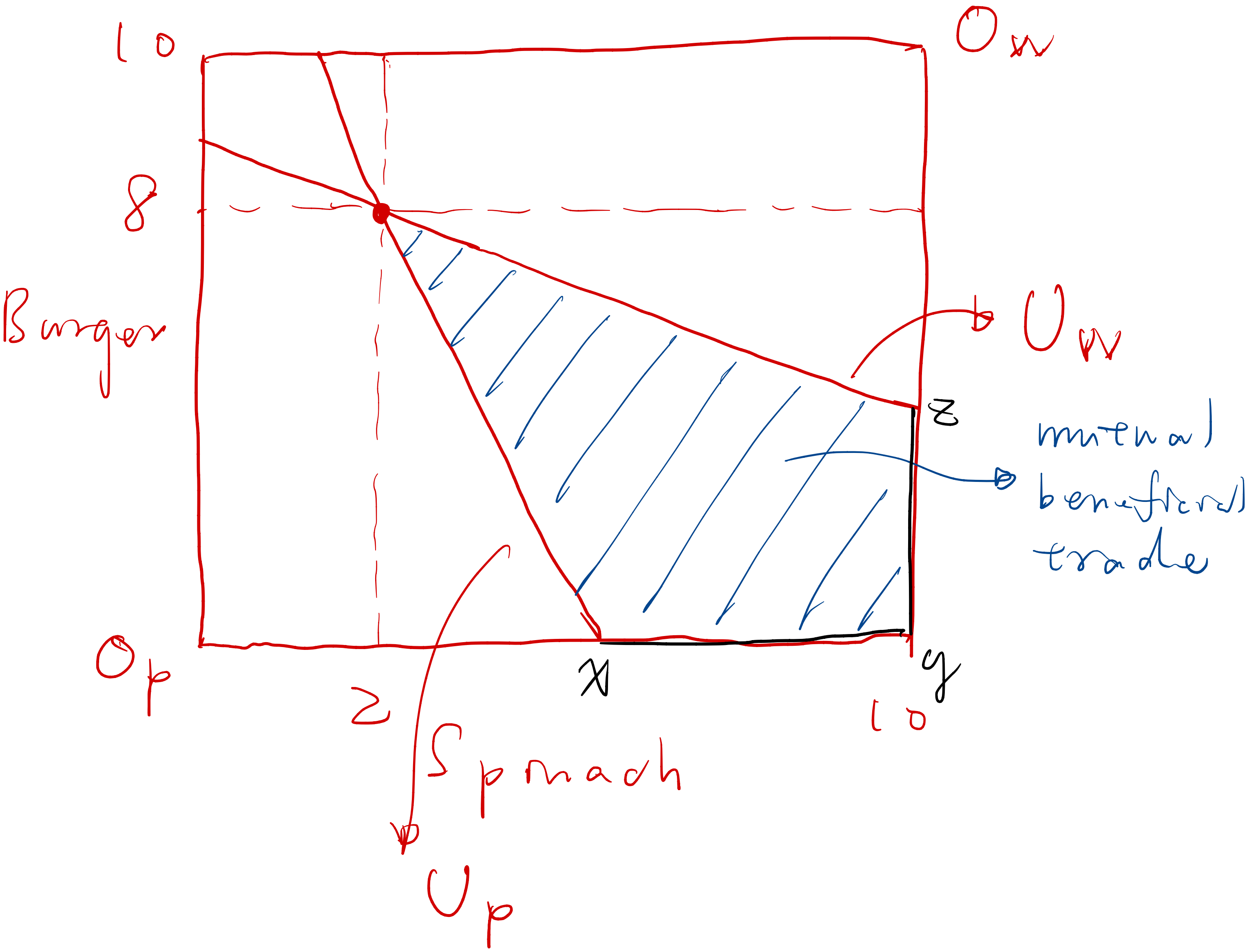
Any allocation on $\overline{xy}$ or $\overline{yz}$ is P.E.

b. [10 points] Popeye and Wimpy trade only with each other. Popeye has 8 hamburgers and 2 cans of spinach, and Wimpy has 2 hamburgers and 8 cans of spinach. Their indifference, somewhat unusually, are all straight lines, Popeye's being much steeper than Wimpy's:



Popeye's indifference curves **Wimpy's indifference curves**

In an Edgeworth box, show the initial endowment, the region of mutual beneficial trade and the pareto efficient allocations.



6. [20 points] (Monopoly) Taipower produces electricity as a natural monopoly and faces market demand given by $Q=12-0.1p$, where Q is in thousands of kilowatt-hours and P is in dollars per kilowatt-hour. Cost function for the electricity company is $C=100+10Q$.

a. [8 points] Solve for the profit maximizing quantity and price. Check if the second order condition is met.

$$\boldsymbol{Q = 12 - 0.1P \rightarrow P = 120 - 10Q \rightarrow MR = 120 - 2(10)Q = 120 - 20Q}$$

$$\boldsymbol{MR = MC \rightarrow 120 - 20Q_M = 10}$$

$$\boldsymbol{S.O.C: dMR/dQ\text{-}dMC/dQ = -20<0}$$

$$\boldsymbol{\rightarrow Q_M = 5.5 \text{ thousand kwh}}$$

$$\boldsymbol{\rightarrow P_M = 120 - 10(5.5) = \$65/\text{kwh}}$$

b. [7 points] Find the elasticity of the demand curve in equilibrium and calculate the Lerner index.

The elasticity of the demand=dQ/dP × P/Q=-0.1 × 65/5.5≅-1.18

The Lerner index= 1/The elasticity of the demand≅0.85

c. [5 points] Should the monopolist shut down or operate? Explain.

p=65>AVC=10 ➔ Operate

7. [10 points] (Price Discrimination) Suppose a railroad faces different demand curves for transporting coal and grain. For coal, $Pc = 38 - Qc$, where Qc is the amount of coal moved when the transport price for coal is Pc. For grain, $Pg = 14 - 0.25Qg$, where Qg is the amount of grain shipped when the transport price for grain is Pg. The marginal cost for moving either commodity is $10. Assuming price discrimination is possible, find the profit-maximizing prices and quantities for coal and grain transport.

   For coal, the marginal revenue curve is $MR_c=38-2Q_c$. Now we equate marginal revenue to marginal cost: $38- 2Q_c=10$, or $Q_c= 14$. Substituting this into the equation for the demand curve, we find: $P_c= 38- 14= 24$. The profit-maximizing rate for transporting coal is $24 per ton-mile.

   For grain, the marginal revenue curve is $MR_g=14- 0.5Q_g$. Now we equate marginal revenue to marginal cost: $14- 0.5Q_g= 10$, or $Q_g= 8$. Substituting this into the equation for the demand curve, we find: $P_g =14 -0.25(8) = 12$. The profit-maximizing rate for transporting grain is *$12* per ton-mile.